BIOCHEMISTRY CUME
FEBRUARY 7, 2003

NADH-Q oxidoreductase (Complex I) is an integral mitochondrial protein which transfers 2 electrons from NADH, with a redox potential of –0.32 V, to coenzyme Q, with a redox potential of +0.1 V. This electron transfer process is coupled to the translocation of 4 protons from the inside to the outside of the mitochondrial membrane. Complex I contains flavin mononucleotide (FMN) and a number of different iron-sulfur centers. The electrical potential across the mitochondrial membrane was measured to be $V_{out} - V_{in} = +0.15$ V, and the pH inside the mitochondrial membrane was $pH_{in} = 8$ and $pH_{out} = 6.5$.

1. (20) Give a simple diagram showing how complex I couples the energy of electron transfer to pumping protons across the mitochondrial membrane. What is known about the mechanism of this process?

2. (25) Calculate the free energy $\Delta G^{o'}$ for electron transfer from 1 mole of NADH to 1 mole of Q.

3. (30) Give a formula for the free energy change ΔG for pumping protons from the inside to the outside of the mitochondrial membrane, and calculate ΔG for proton pumping by complex I under the above conditions.

4. (25) Calculate the efficiency with which complex I couples the exergonic free energy of electron transfer to the endergonic free energy of proton pumping.

Possibly useful formulas:

$\Delta G^{o'} = -RT\ln K_{eq}$; $\Delta G^{o} = -nF\Delta E_{o}$
$\Delta G = \Delta G^{o} + RT\ln\{\text{products/reactants}\} + nF\Delta V$
F = Faraday's constant = 23 kcal/volt; $R = 1.987 \text{cal mol}^{-1} \text{deg}^{-1}$